

Literature-Implied Answer: Convex Weak-Star Derived Sets Above $\omega + 1$

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Status. literature_implied_answer (source Question 2; source Question 1 remains open).
This is a literature-status packet, not an original proof from the run.

Source Question

The source paper is Zdenek Silber, *Weak* derived sets of convex sets in duals of non-reflexive spaces*, arXiv:2101.10684, J. Funct. Anal. 281 (2021), no. 12, Paper No. 109259 [1]. In Section 3, “Remarks and open problems”, Silber asks two questions. The second asks:

if X is non-reflexive and quasi-reflexive, are there convex subsets of X^* with order higher than $\omega+1$?

Supporting Theorem

The supporting paper is Mikhail I. Ostrovskii, *Weak* closures and derived sets for convex sets in dual Banach spaces*, arXiv:2112.04670 [2]. The abstract and Theorem 1 state the successor-ordinal result for convex subsets of duals: for every non-reflexive Banach space and every countable successor ordinal, there is a convex subset of the dual whose weak-star derived-set process reaches the weak-star closure precisely at that successor stage.

Identification

Ostrovskii’s theorem applies to every non-reflexive Banach space, hence in particular to every non-reflexive quasi-reflexive Banach space. Choosing any countable successor ordinal strictly larger than $\omega + 1$, for instance $\omega + 2$, gives a convex subset of X^* with order higher than $\omega + 1$. Therefore Silber’s second question has a positive answer in the later literature.

The supporting paper cites Silber’s 2021 paper and presents its theorem as a further development of the convex-set weak-star derived-set theory. The answer to the quasi-reflexive question is therefore immediate from the theorem, but the supporting paper does not appear to restate that specific question verbatim; this is why the packet is placed under `literature_implied_answers`.

Limitations

This packet does not answer Silber’s first question asking whether the order of a convex subset of the dual of a separable Banach space can be a limit ordinal. Ostrovskii’s paper explicitly records that this limit-order question remains open in its final comment section.

Search and Provenance

The cheap run indexes were searched for arXiv:2101.10684, weak derived sets, derived sets, convex sets, and related phrases. No prior packet or attempt was found. Local source search and web search then identified arXiv:2112.04670 as the decisive later theorem.

References

- [1] Z. Silber, *Weak* derived sets of convex sets in duals of non-reflexive spaces*, J. Funct. Anal. 281 (2021), no. 12, Paper No. 109259; arXiv:2101.10684.
- [2] M. I. Ostrovskii, *Weak* closures and derived sets for convex sets in dual Banach spaces*, arXiv:2112.04670.